DA

What Developers Use and Want

Stack Overflow Developer Survey 2024 and 27,005 job postings

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github.com/Sumant-crty/IBM-Data-Analyst-Capstone-Project

Three data sources, four questions

65,437

survey responses (Stack Overflow 2024)

185

countries represented

27,005

job postings from a JSON API

1. Which technologies do developers use today?
2. Which do they want to use next, and where is the gap biggest?
3. Who are the respondents?
4. Which skills appear most in job postings?

From raw survey and API data to a dashboard

1

Collect

Survey CSV, jobs JSON API, salary table scraped from a web page

requests, BeautifulSoup

2

Clean

Removed duplicates, imputed missing values, normalized compensation

pandas

3

Explore

Split multi-select answers; distributions, outliers, correlations

pandas, Seaborn

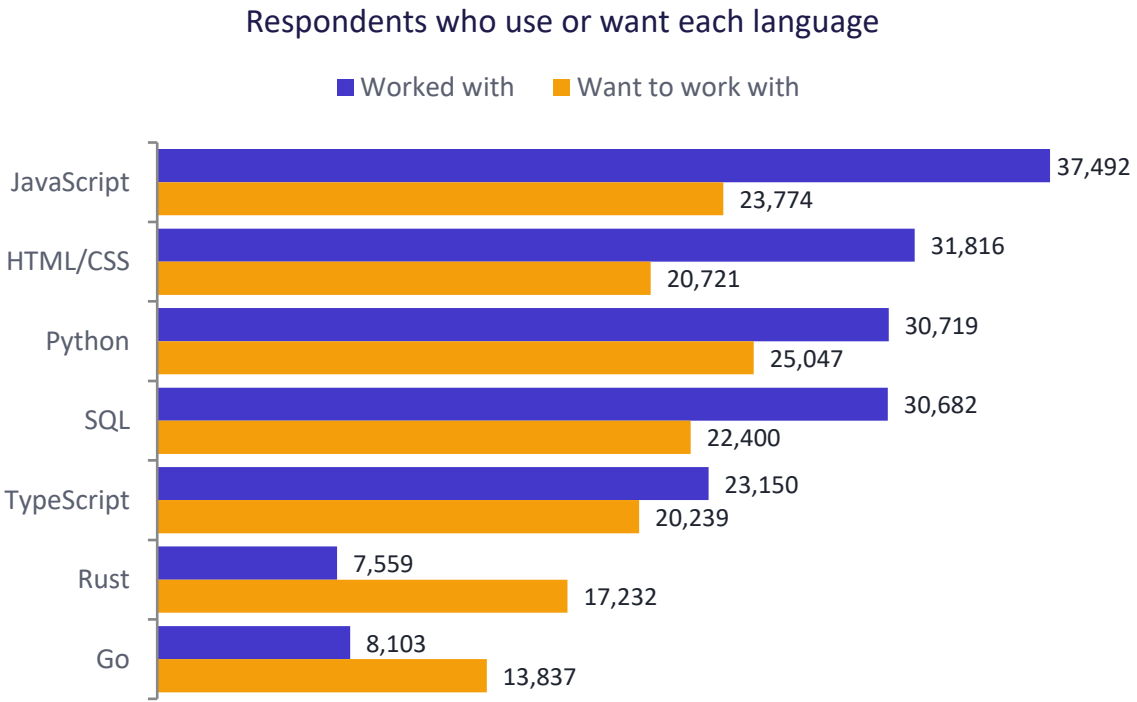
4

Visualize

Three-tab dashboard: current tech, future tech, demographics

IBM Cognos Analytics

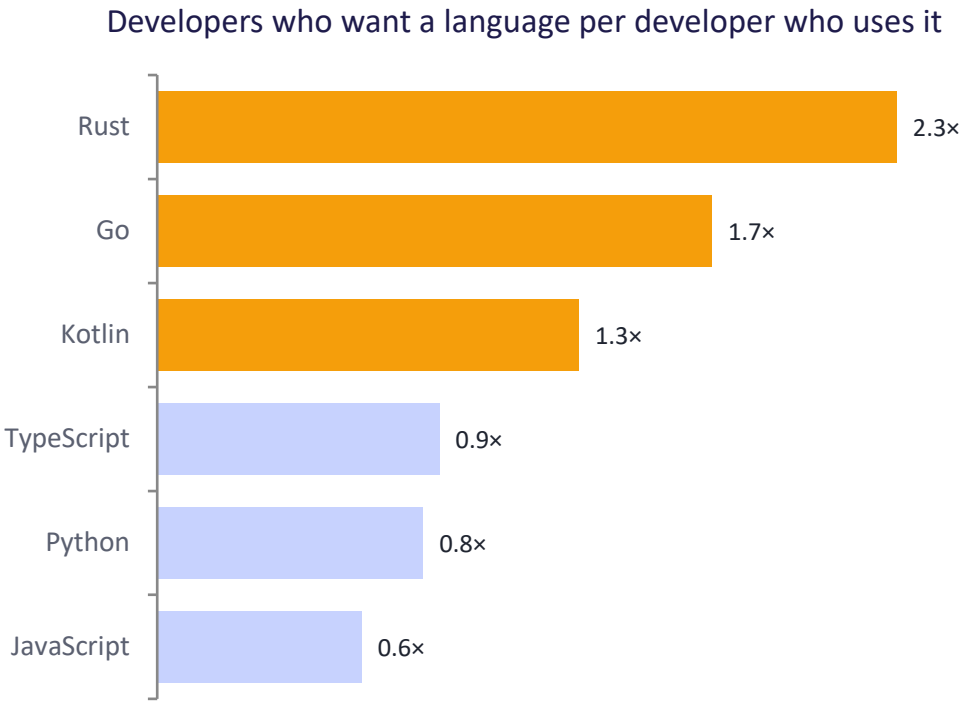
JavaScript is used most; Python is wanted most



Python is the only top language that more developers want than JavaScript.
SQL is used by half of all respondents who answered: it is the most common data skill.

Counts are respondents; each person could pick several languages.

Rust and Go are wanted far more than they are used



17,232

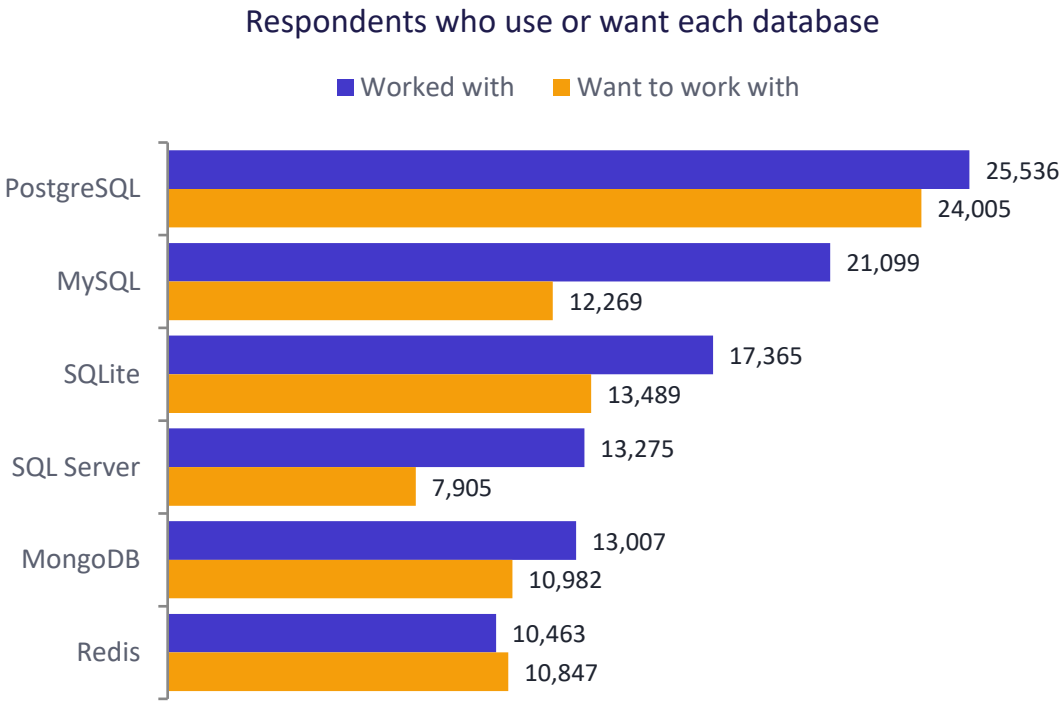
want Rust, but only 7,559 use it

13,837

want Go, against 8,103 who use it

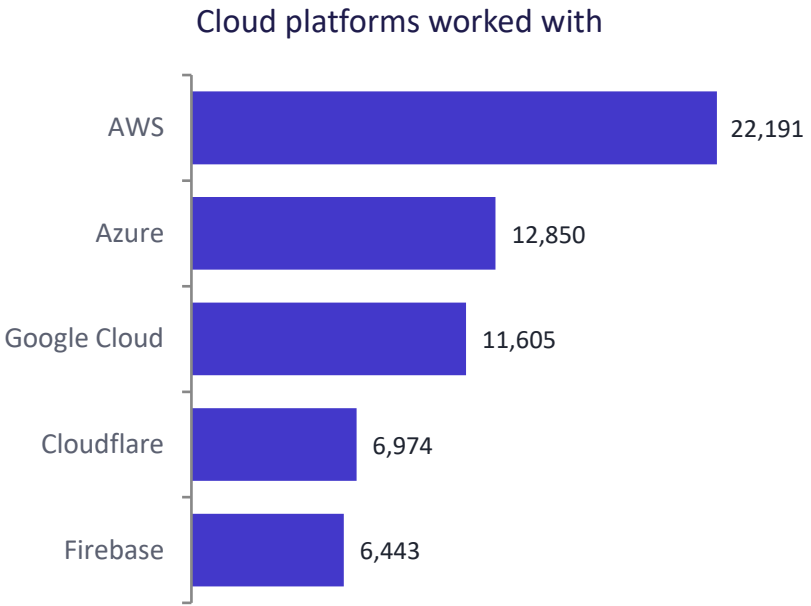
Rust and Go users also report the highest median pay in the survey (about $76K).

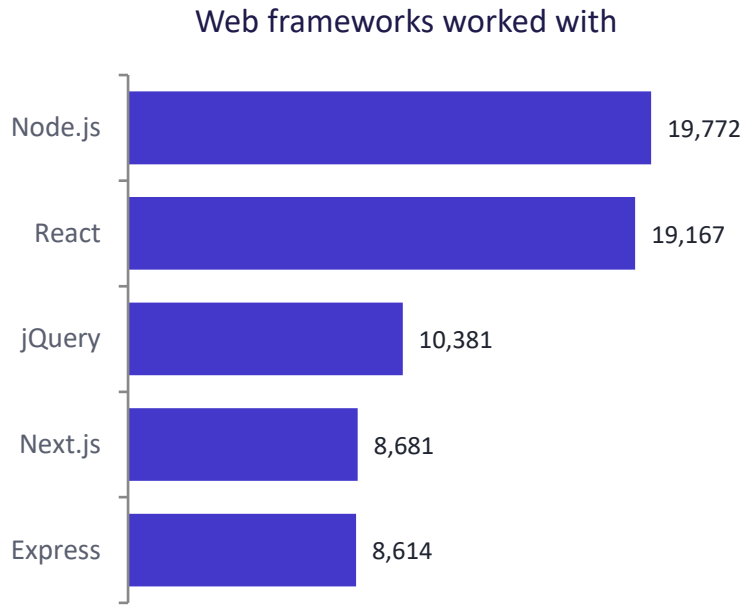
PostgreSQL leads both today and next year



PostgreSQL keeps almost all its users' interest (24,005 want it vs 25,536 using it).
Redis is the only major database more developers want than use.
MySQL and SQL Server lose the most interest.

AWS and the React/Node.js stack dominate





AWS is used by almost as many respondents as Azure and Google Cloud combined. React, Node.js and Next.js also lead future interest.

Who answered the survey

58%

are aged 18–34

72%

hold a bachelor's degree or higher

7 yrs

median professional coding experience

$65K

median yearly pay (23,435 who reported it)

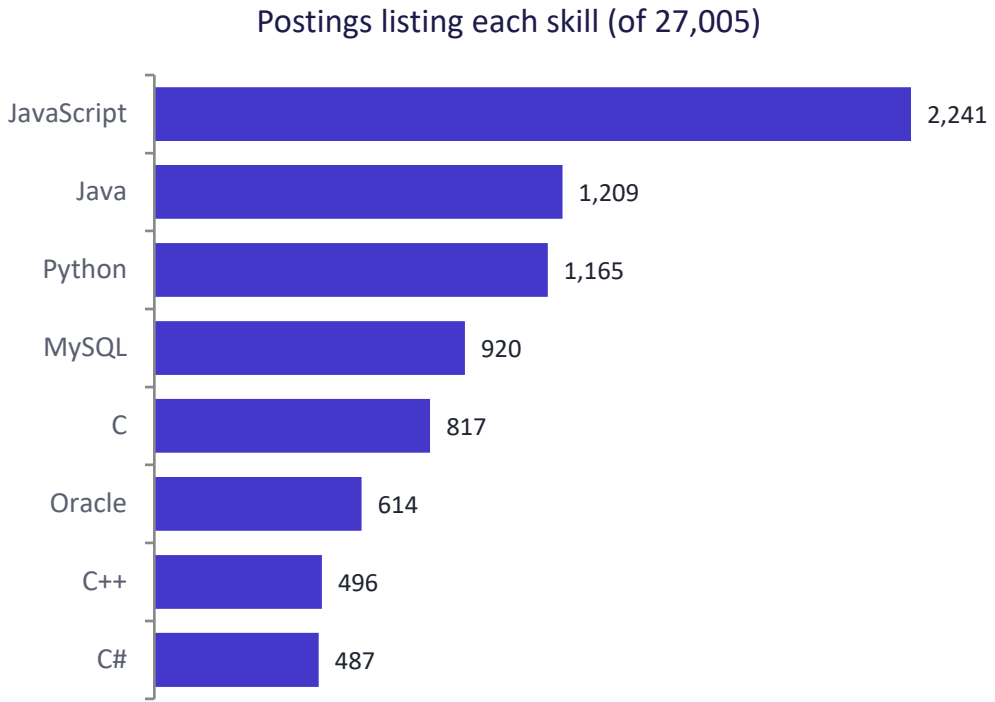
6.9 / 10

average job satisfaction

US · DE · IN

most responses: 11,095 · 4,947 · 4,231

JavaScript, Java and Python lead job postings



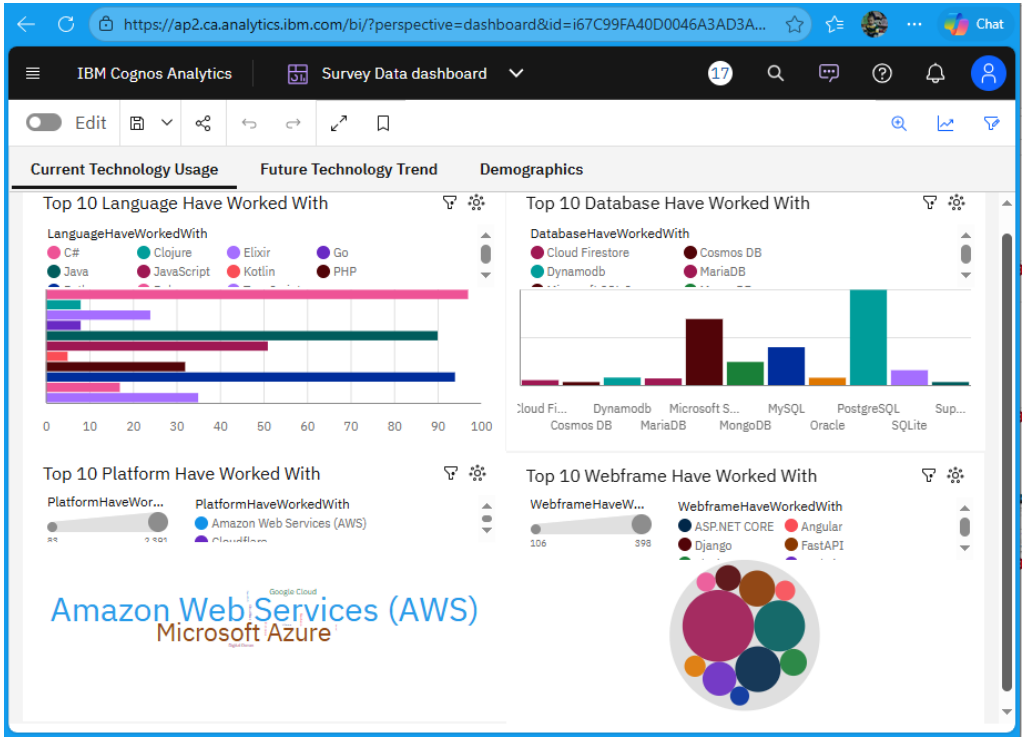
Data-quality catch

The first count matched skills by substring, so "C" matched any skill containing the letter c.

C: 25,114 → 817
Java: 3,428 → 1,209

Matching whole skill names fixed the ranking.

An interactive dashboard in IBM Cognos Analytics



Three tabs

- Current technology usage
- Future technology trends
- Demographics: age, country, education

Built on a cleaned subset of the survey, so its counts are lower than the full-survey numbers in this deck.

Key takeaways

1. **Core stack** JavaScript/TypeScript, Python and SQL with PostgreSQL cover most of what developers use

2. **Rising** Rust and Go have far more developers wanting them than using them

3. **Cloud** AWS leads; Azure and Google Cloud follow

4. **Caveat** Survey respondents are self-selected Stack Overflow users; "want" is intent, not adoption

Code, data notes and dashboard: github.com/Sumant-crty/IBM-Data-Analyst-Capstone-Project